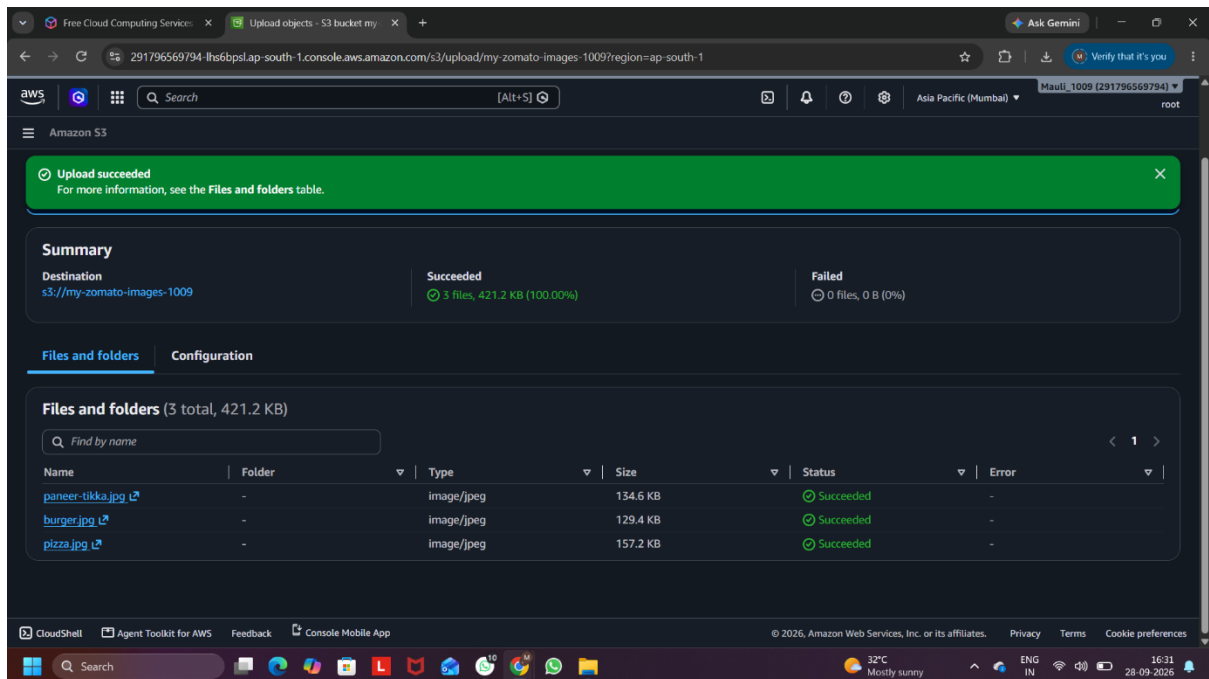


Cloud-Storage-Assignment-3

Q1. Create a new S3 bucket named 'my-zomato-images' and upload at least 3 sample image files representing restaurant menu items.

Ans:



Screenshot 1: Successfully Created the S3 Bucket and Uploaded Three Restaurant Menu Images (Pizza, Burger, and Paneer Tikka).

Result: The S3 bucket was created successfully, and three restaurant menu images were uploaded.

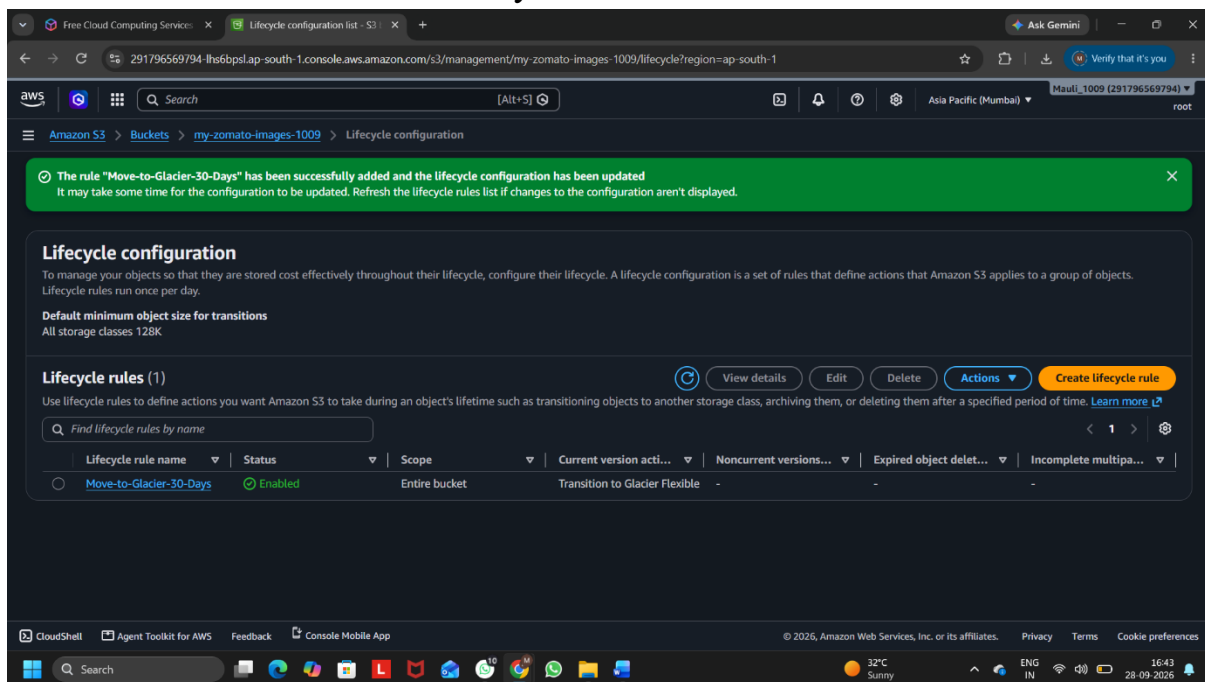
Q2. Set up a lifecycle policy on your 'my-zomato-images' S3 bucket to automatically transition all objects older than 30 days to the Glacier storage class. Hint: Use the S3 Management Console's 'Lifecycle rules' section to define the transition action.

Ans:

Steps:

1. Opened the S3 bucket and navigated to the Management tab.
2. Created a lifecycle rule to apply to all objects in the bucket.
3. Configured the rule to transition objects to Glacier Flexible Retrieval after 30 days.

4. Saved and verified the lifecycle rule.



Screenshot 1: Successfully Created S3 Lifecycle Rule to Transition Objects to Glacier After 30 Days

Result: A lifecycle policy was configured to automatically transition all objects older than 30 days to the S3 Glacier Flexible Retrieval storage class.

Q3. Simulate a cost optimization scenario: calculate and compare the estimated monthly storage cost for keeping 5GB of images in S3 Standard versus S3 Glacier using AWS Pricing Calculator. Hint: Use the AWS Pricing Calculator website and note down the difference in INR.

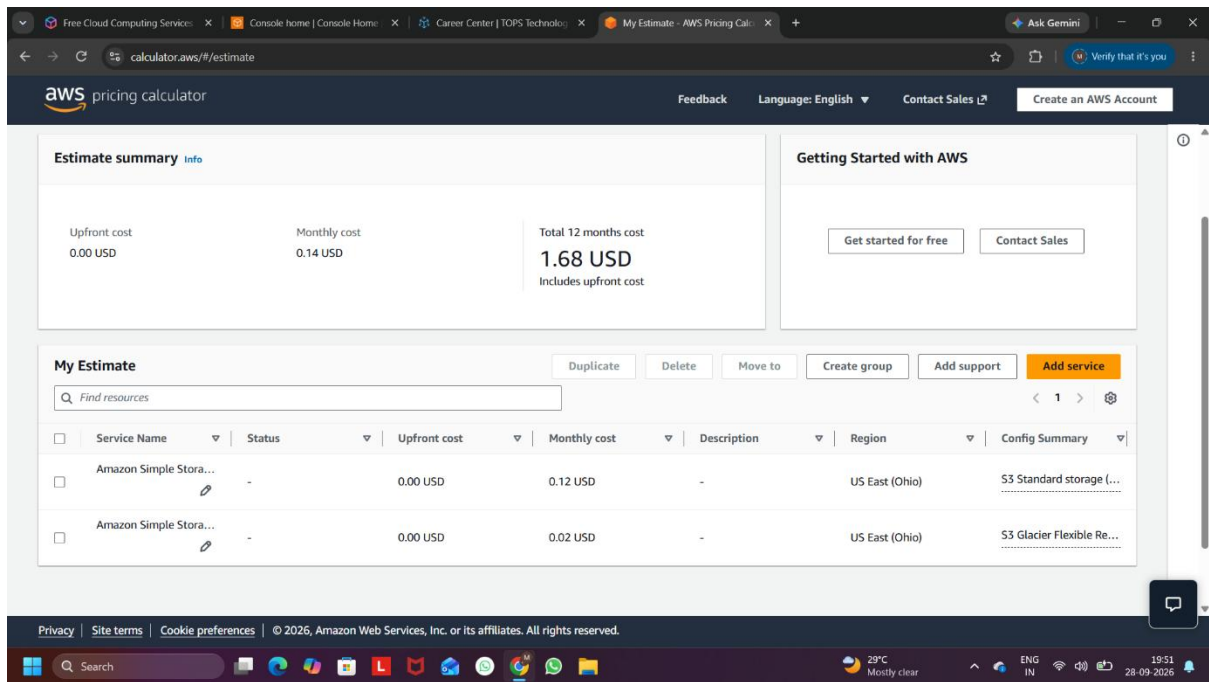
Ans:

WS Pricing Calculator was used to estimate and compare the monthly storage cost of storing **5 GB of images** in Amazon S3 Standard and S3 Glacier Flexible Retrieval.

Storage Class Estimated Monthly Cost

S3 Standard **\$0.12 ≈ ₹11.51**

S3 Glacier **\$0.02 ≈ ₹1.92**

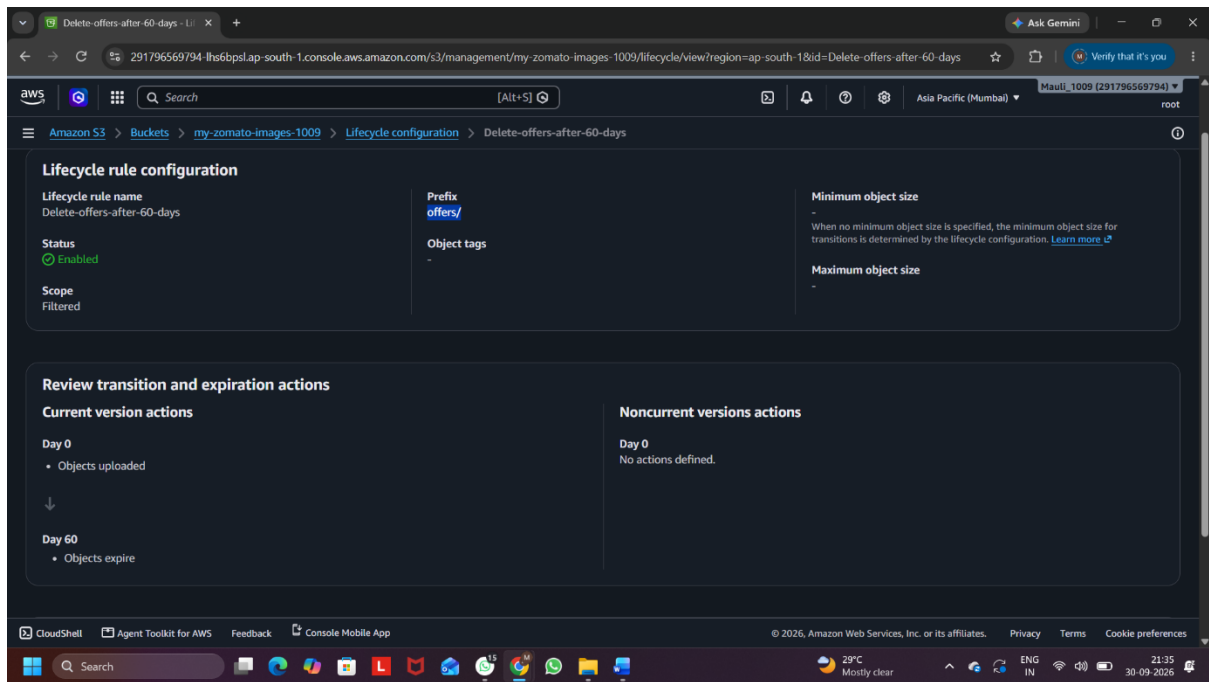


Screenshot 1 : AWS Pricing Calculator showing estimated monthly costs for S3 Standard and S3 Glacier.

Q4. Add a new rule to your S3 bucket's lifecycle policy so that any objects with the prefix 'offers/' (e.g., 'offers/summer2024.jpg') are deleted automatically after 60 days.

Ans:

1. Open the Amazon S3 console and select the required bucket.
2. Go to the Management tab and scroll down to the Lifecycle rules section.
3. Click Create lifecycle rule.
4. Enter the rule name as Delete-offers-after-60-days.
5. Select Limit the scope to specific prefixes or tags and enter offers/ as the prefix.
6. Under Lifecycle rule actions, select the option to expire/delete current versions of objects.
7. Set the expiration period to 60 days.
8. Click Create rule to save the lifecycle rule.



Screenshot 1: Lifecycle rule configured to automatically delete objects with the ‘offers/’ prefix after 60 days.

Q5. Use ChatGPT to generate a sample JSON configuration for an S3 lifecycle policy that transitions files to Glacier after 15 days and deletes them after 180 days. Paste the generated JSON and briefly explain each section.

Ans:

AI-Generated S3 Lifecycle Policy JSON Configuration:

```
{
  "Rules": [
    {
      "ID": "Transition-to-Glacier-and-Delete",
      "Status": "Enabled",
      "Filter": {},
      "Transitions": [
        {
          "Days": 15,
          "StorageClass": "GLACIER"
        }
      ]
    }
  ],
}
```

```
"Expiration": {  
  "Days": 180  
}  
}  
]  
}
```

Brief Explanation

- **Rules:** Contains the lifecycle rules for the S3 bucket.
- **ID:** A unique name used to identify the lifecycle rule.
- **Status:** Enabled means the lifecycle rule is active.
- **Filter:** {} means the rule applies to all objects in the bucket.
- **Transitions:** Moves objects to **S3 Glacier after 15 days**.
- **Expiration:** Automatically **deletes objects after 180 days**.

Conclusion: This lifecycle policy helps reduce storage costs by moving older files to Glacier and automatically removing them after 180 days.